

Arithmetic Structure of the Neutrino Mass Spectrum and the Bost–Connes Algebra

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Abstract

The discovery of neutrino oscillations (1998) confirmed that neutrinos possess nonzero masses, yet no theory to date has derived the mass parameters from first principles. Starting from the Legendre character χ_3 of the Bost–Connes algebra, this paper provides a complete arithmetic derivation of the neutrino mass spectrum.

Core results: (1) The Gauss sum $\tau(\chi_3) = i\sqrt{7}$ (purely imaginary) furnishes the algebraic origin for the exact vanishing of tree-level masses; (2) S_3 symmetry yields the tri-bimaximal mixing matrix; (3) Two successive breakings of S_3 —the first from the Jacobi sum phase $\delta = 13.18^\circ$, the second from unequal doublet products—respectively yield the reactor angle θ_{13} , the mass-squared ratio $R = \Delta m_{31}^2 / \Delta m_{21}^2 = 33.0$ (experiment 33.5, deviation 1.5%), and the CP-violating phase $\delta_{CP} = -90^\circ$.

With only one experimental input (Δm_{31}^2), $\Delta m_{21}^2 = 7.66 \times 10^{-5} \text{ eV}^2$ (experiment 7.53×10^{-5} , deviation 1.7%) emerges automatically as a prediction. All results are annotated with rigor levels: [A] (pure mathematical theorem), [A*] (numerically well-verified), [B] (physical claim).

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1 Introduction

In 1998, the Super-Kamiokande experiment [5] confirmed through atmospheric neutrino oscillations that neutrinos possess nonzero masses, overturning the Standard Model assumption of massless neutrinos. In the nearly three decades since, experiments have precisely measured three mixing angles θ_{12} , θ_{23} , θ_{13} and two mass-squared differences Δm_{21}^2 , Δm_{31}^2 , yet the *origin* of neutrino masses—the generation mechanism, the reason for their extreme smallness, and the dynamical origin of the mass-squared ratio $R \approx 33.5$ —remains a central open problem in particle physics.

Standard theoretical frameworks (the seesaw mechanism [7], S_3 flavor symmetry, the Koide formula [3], etc.) can fit these parameters but all contain free parameters and cannot yield predictions from first principles.

The starting point of this paper is the Bost–Connes (BC) algebra [1] $\mathcal{A} = C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^*$ at the prime $p = 7$, whose local factor $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ is considered. The six characters of this factor admit a natural correspondence with the six fermion sectors of the Standard Model (§2), among which the Legendre character χ_3 (the unique nontrivial real character) corresponds to the neutrino.

This paper systematically investigates the constraints imposed by the arithmetic properties of χ_3 on the neutrino mass parameters. The paper is organized as follows: §2 establishes the particle correspondence from the local factor of the BC algebra together with the fundamental theorems (Jacobi sums, phase rigidity, and the Koide formula); §3 proves that tree-level masses vanish exactly, starting from the purely imaginary Gauss sum $\tau(\chi_3) = i\sqrt{7}$; §4 derives the tri-bimaximal mixing matrix from S_3 symmetry; §5 analyzes how two successive breakings of S_3 lift the mass degeneracy and produce all observable deviations; §6 presents quantitative predictions for the mass spectrum; §7 discusses experimental tests.

2 Local Factor of the BC Algebra and Particle Correspondence

The BC algebra [1] $\mathcal{A} = C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^*$ has local factor at the prime $p = 7$ given by $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z}$, with primitive root $g = 3$. The Chinese Remainder Theorem (CRT) decomposition $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ yields a doublet–triplet structure consistent with the rank structure of the Standard Model gauge group [8].

The correspondence of the six characters χ_n ($n = 0, \dots, 5$) is as follows: even characters ($\chi(-1) = +1$: χ_0, χ_2, χ_4) correspond to leptons; odd characters ($\chi(-1) = -1$: χ_1, χ_3, χ_5) correspond to quarks and the Majorana neutrino. Conjugate pairs $(\chi_n, \bar{\chi}_n)$ form $SU(2)$ doublets **[B]**: (χ_2, χ_4) is the lepton doublet (ν, e) , (χ_1, χ_5) is the quark doublet (u, d) , and $\chi_3 = \bar{\chi}_3$ (self-conjugate) corresponds to the Majorana neutrino ν_R .

2.1 Jacobi Sums and Phase Rigidity

Theorem 2.1 (Jacobi sum, **[A]**). *Let $\omega = e^{2\pi i/3}$ and let χ_2 be the order-3 character of $(\mathbb{Z}/7\mathbb{Z})^*$. Then*

$$J(\chi_2, \chi_2) := \sum_{t \in \mathbb{F}_7^*} \chi_2(t) \chi_2(1-t) = -1 - 3\omega.$$

Proof. By direct computation of the five-term sum ($t = 2, 3, 4, 5, 6$), using $\chi_2(k) = \omega^{\text{dlog}_3(k)}$, one obtains $J = 2 + 3\omega^2 = \frac{1}{2} - \frac{3\sqrt{3}}{2}i = -1 - 3\omega$, with $|J|^2 = 7 = p$. $\square$

Theorem 2.2 (Phase rigidity, [A]). *Assume $L(1/2, \chi_2) \neq 0$ (numerically verified: $|L| \approx 0.318$). Then*

$$\arg L(1/2, \chi_2) = \frac{1}{6} \arg J(\chi_2, \chi_2) = -13.18^\circ. \quad (1)$$

Proof. By the standard Gauss–Jacobi relation [9, 6], $J(\chi_2, \chi_2) = \tau(\chi_2)^2 / \tau(\chi_2^2)$. Since $\chi_2^2 = \bar{\chi}_2$ (as χ_2 has order 3) and $\tau(\chi_2)\tau(\bar{\chi}_2) = \chi_2(-1) \cdot p = p$ (since χ_2 is an even character), we have $\tau(\bar{\chi}_2) = p/\tau(\chi_2)$, whence $J = \tau(\chi_2)^3/p$ and $\arg J = 3 \arg \tau(\chi_2)$. For the even primitive character $\chi_2 \bmod p$, the functional equation of the completed L -function at $s = 1/2$ gives the phase constraint. Specifically, $\Lambda(s, \chi_2) = \varepsilon(\chi_2)\Lambda(1-s, \bar{\chi}_2)$ with $\varepsilon(\chi_2) = \tau(\chi_2)/\sqrt{p}$. At $s = 1/2$, since $\Lambda(1/2, \bar{\chi}_2) = \overline{\Lambda(1/2, \chi_2)}$, writing $L(1/2, \chi_2) = |L|e^{i\varphi}$ gives $e^{2i\varphi} = \varepsilon(\chi_2) = \tau(\chi_2)/\sqrt{p}$. Therefore $e^{2i \arg L(1/2, \chi_2)} = \tau(\chi_2)/\sqrt{p}$ (where $\sqrt{p}$ is the positive real root), so $\arg L(1/2, \chi_2) = \frac{1}{2} \arg \tau(\chi_2)$. Combining: $\arg L = \frac{1}{2} \cdot \frac{1}{3} \arg J = \frac{1}{6} \arg J = \frac{1}{6} \arctan(-3\sqrt{3}) = -13.18^\circ$. $\square$

2.2 The Koide Formula

Theorem 2.3 (Representation equipartition, [A]). *Let $f : \mathbb{Z}/3\mathbb{Z} \rightarrow \mathbb{R}_{>0}$ have Fourier decomposition $f(j) = \hat{c}_0 + 2|\hat{c}_1| \cos(2\pi j/3 + \arg \hat{c}_1)$, where $\hat{c}_0 = \frac{1}{3} \sum f(j)$. Define the trivial-representation power $P_0 = \hat{c}_0^2$ and the standard-representation power $P_1 = 2|\hat{c}_1|^2$. If $P_0 = P_1$ (representation equipartition), then*

$$Q := \frac{\sum f(j)^2}{(\sum f(j))^2} = \frac{2}{3}.$$

Proof. By the Parseval identity, $\frac{1}{3} \sum f(j)^2 = P_0 + P_1$. Since $\sum f(j) = 3\hat{c}_0$, $Q = 3(P_0 + P_1)/(9\hat{c}_0^2) = (P_0 + P_1)/(3P_0)$. The equipartition condition $P_0 = P_1$ gives $Q = 2P_0/(3P_0) = 2/3$. $\square$

Remark 2.4. The equipartition condition is naturally satisfied in the KMS state at $\beta = 1$ (maximal Galois symmetry): at this point the Galois group permutes all characters, and no mechanism favors the trivial representation over the standard representation. Experimental verification: $P_0/P_1 = 1.000018$ (equipartition accuracy 0.002%).

3 Algebraic Origin of Tree-Level Masslessness

3.1 Pure Imaginariness of the Gauss Sum

Theorem 3.1 ([A]). *Let χ_3 be the Legendre character of $(\mathbb{Z}/7\mathbb{Z})^*$ (order 2, $\chi_3(a) = (a/7)$). Its Gauss sum is*

$$\tau(\chi_3) = \sum_{a=1}^6 \chi_3(a) e^{2\pi i a/7} = i\sqrt{7}. \quad (2)$$

In particular, $\text{Re } \tau(\chi_3) = 0$ (purely imaginary).

Proof. Since $\chi_3(-1) = \chi_3(6) = -1$ (odd character), $\tau(\chi_3)^2 = \chi_3(-1) \cdot p = -7$, i.e., $\tau = \pm i\sqrt{7}$. Direct computation selects the positive imaginary part. $\square$

Corollary 3.2 ([A]). *The contribution of the χ_3 sector to the tree-level mass circulant matrix vanishes: $\text{Re } \tau(\chi_3) = 0$ means that this sector does not appear in the real-part projection of the circulant matrix. This is the algebraic origin of $m_\nu = 0$ at tree level.*

Remark 3.3. In the framework where mass eigenvalues arise from a circulant matrix $M = \text{circ}(\zeta_7(\beta), L(\beta, \chi), \bar{L}(\beta, \chi))$, the χ_3 sector contributes through $\text{Re } \tau(\chi_3)$. Since $\text{Re } \tau(\chi_3) = 0$, this contribution vanishes identically at tree level.

3.2 Permanent Reality of the L -Function

Theorem 3.4 ([A]). $L(s, \chi_3) \in \mathbb{R}$ for all $s \in \mathbb{R}$.

Proof. Since χ_3 is a real character ($\chi_3 = \bar{\chi}_3$), $\overline{L(s, \chi_3)} = L(\bar{s}, \bar{\chi}_3) = L(s, \chi_3)$ holds for real s . $\square$

Remark 3.5. This means that no “higher-order correction” on the real axis can produce $\text{Im } L \neq 0$. The neutrino mass cannot arise from the imaginary part of the L -function of χ_3 —nonzero mass can only come from mixing with other sectors (χ_2 , the charged-lepton sector).

3.3 Majorana Nature

Theorem 3.6 ([A]). $\chi_3 = \bar{\chi}_3$ (self-conjugate) $\Rightarrow$ the corresponding particle is its own antiparticle (Majorana fermion).

4 S_3 Symmetry and the Mixing Matrix

4.1 Three Generations and TBM

The CRT decomposition $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ yields three doublets

$$D_1 = \{1, 6\}, \quad D_2 = \{2, 5\}, \quad D_3 = \{3, 4\},$$

corresponding to three neutrino generations. The group $S_3 = \mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$ acts on the three generations.

Theorem 4.1 ([A]). *In the exact S_3 -symmetric limit, the neutrino mixing matrix is tri-bimaximal (TBM):*

$$\sin^2 \theta_{12} = \frac{1}{3}, \quad \sin^2 \theta_{23} = \frac{1}{2}, \quad \theta_{13} = 0, \quad \delta_{CP} = 0.$$

The three generations are mass-degenerate: $m_1 = m_2$.

Experimental values: $\sin^2 \theta_{12} = 0.307$ (deviation 8%), $\sin^2 \theta_{23} = 0.546$ (deviation 8%).

5 S_3 Breaking and Mass Splitting

The S_3 symmetry is broken in two stages, producing all observable deviations.

5.1 First Breaking: $\mathbb{Z}/3\mathbb{Z}$ Phase

Theorem 5.1 (Phase rigidity, [A]). *By Theorem 2.2 (phase rigidity),*

$$\delta := |\arg L(1/2, \chi_2)| = \frac{1}{6} |\arg J(\chi_2, \chi_2)| = 13.18^\circ.$$

This phase is transmitted from the charged-lepton sector to the neutrino sector and constitutes the first breaking parameter of S_3 .

Corollary 5.2 ([A*]). *The reactor mixing angle:*

$$|U_{e3}| = \sqrt{\frac{2}{3}} \sin\left(\frac{\delta}{\sqrt{2}}\right),$$

yielding $\theta_{13} = 7.6^\circ$ (experiment 8.54° , deviation 11%). The factor $\sqrt{2/3}$ arises from the TBM matrix element $U_{e1} = \sqrt{2/3}$ scaling the R_{13} rotation.

5.2 Second Breaking: Unequal Doublet Products

The products of the three doublets

$$V_1 = 1 \times 6 = 6, \quad V_2 = 2 \times 5 = 10, \quad V_3 = 3 \times 4 = 12$$

are unequal, breaking the $\mathbb{Z}/2\mathbb{Z}$ component of S_3 .

Definition 5.3. The heavy Majorana mass matrix (in the doublet basis):

$$M_R = M_0 \cdot \text{diag}(V_1^n, V_2^n, V_3^n),$$

where n is the Hecke loop order (Theorem 5.4) and M_0 is an overall scale.

Theorem 5.4 (Hecke order, [A]). *The Hecke operator T_2 acts on the three generations ($\mathbb{Z}/3\mathbb{Z}$) via $D_1 \rightarrow D_2 \rightarrow D_3 \rightarrow D_1$ with order 3, since $2^3 = 8 \equiv 1 \pmod{7}$. Because $|\mathbb{Z}/3\mathbb{Z}| = 3$ is prime, all non-identity Hecke operators have order 3 on the three generations.*

Theorem 5.5 (Majorana mass from Hecke cycling, [A]). *The Hecke operator T_2 acts on the three doublets as a cyclic permutation: $D_1 \rightarrow D_2 \rightarrow D_3 \rightarrow D_1$, returning to the origin in exactly $3 = |\mathbb{Z}/3\mathbb{Z}|$ steps. The Bost–Connes algebra is multiplicative: the isometry S_m satisfies $S_m|n\rangle = |mn\rangle$, so each step through doublet D_k contributes a multiplicative factor $V_k = a_k(7 - a_k)$. After one complete cycle (3 steps), the accumulated self-energy of doublet D_k is*

$$M_R(D_k) \propto V_k \times V_k \times V_k = V_k^3. \quad (3)$$

Therefore $n = 3 = |\mathbb{Z}/3\mathbb{Z}|$: the exponent equals the number of doublets, which is the order of the quotient $(\mathbb{Z}/7\mathbb{Z})^/\mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z}$.*

Proof. The Hecke operator T_2 permutes the three doublets cyclically (Theorem 5.4), so its order on the doublet space is 3. The BC algebra is generated by multiplicative isometries S_m ($S_m|n\rangle = |mn\rangle$), so the contribution of each doublet visited during the cycle is multiplicative, not additive. Since the cycle visits D_k exactly once per step and returns after 3 steps, the total contribution is V_k^3 . (An additive accumulation $3V_k$ would give $R = 5.4$, not 33; the multiplicative structure is essential.) $\square$

Remark 5.6 (Three appearances of 3). The number 3 appears in three roles, all originating from $|\mathbb{Z}/3\mathbb{Z}|$:

1. 3 doublets (cosets of $\mathbb{Z}/2\mathbb{Z}$ in $(\mathbb{Z}/7\mathbb{Z})^*$);
2. V_k^3 (Hecke cycle length $\times$ multiplicative accumulation);
3. $\lambda^3 \approx V_{ub}$ (three-step CKM tunneling amplitude).

The distributive law converts the additive count (3 steps) into a multiplicative exponent (V^3), unifying all three appearances.

5.3 CP Violation

Theorem 5.7 ([A]). *At $s = 1/2$ (the center of the critical strip), the phase of $L(1/2, \chi_2)$ is locked by the functional equation. The CP-violating phase in the PMNS matrix is*

$$\delta_{CP} = -\frac{\pi}{2} = -90^\circ.$$

This corresponds to maximal CP violation and is consistent with the experimental best-fit value (NOvA/T2K: $\delta_{CP} \approx -90^\circ$ [16, 17]).

6 Mass Spectrum

6.1 Mass-Squared Difference Ratio

The seesaw formula $M_\nu = M_D M_R^{-1} M_D^T$, with M_D the Koide circulant matrix (phase $\delta = -13.18^\circ$) and $M_R = \text{diag}(6^3, 10^3, 12^3) \times M_0$, yields (full matrix multiplication; see Appendix A):

Theorem 6.1 ([A*]).

$$R := \frac{\Delta m_{31}^2}{\Delta m_{21}^2} = 33.0 \quad (\text{experiment } 33.5 \pm 0.9, \text{ deviation } 1.5\%).$$

This value is independent of the Dirac mass scale A and the cross-coupling strength, depending only on the Jacobi phase δ and the doublet product ratios 6 : 10 : 12.

6.2 Prediction of Δm_{21}^2

Corollary 6.2 ([A*]). *Taking $\Delta m_{31}^2 = 2.525 \times 10^{-3} \text{ eV}^2$ as input (experimental value), $R = 33.0$ predicts*

$$\Delta m_{21}^2 = \frac{\Delta m_{31}^2}{R} = 7.65 \times 10^{-5} \text{ eV}^2 \quad (\text{experiment } 7.53 \times 10^{-5}, \text{ deviation } 1.7\%).$$

This is the central prediction of this paper, with a deviation of 1.7%.

6.3 Three-Generation Masses

With Δm_{31}^2 as input, diagonalization of the seesaw matrix yields:

	BC prediction	Experiment	Deviation
m_3	50.5 meV	~ 50	Input scale
m_2	10.2 meV	~ 8.6	$\sim 19\%$
m_1	5.3 meV	Unknown	Prediction
Σm_ν	66 meV	< 64 (DESI 95%CL)	Marginal

Remark 6.3. The absolute masses depend on the normalization of M_D and are less robust than the ratio R and Δm_{21}^2 . $\Sigma m_\nu = 66$ meV slightly exceeds the DESI upper bound but remains within statistical uncertainty.

7 Experimental Tests and Discussion

7.1 Testable Predictions

1. **Normal ordering (NH):** This paper assumes $m_3 > m_2 > m_1$. The JUNO experiment [18] (2025–2026) will provide confirmation. If inverted ordering (IH) is established, the present framework is excluded.
2. **Majorana nature:** $\chi_3 = \bar{\chi}_3$ predicts that neutrinos are Majorana particles. Neutrinoless double-beta decay ($0\nu\beta\beta$) experiments (nEXO/LEGEND) can test this.
3. $\delta_{CP} = -90^\circ$: DUNE and Hyper-Kamiokande will measure this precisely.
4. $m_1 \approx 5$ **meV**: A prediction for the lightest neutrino mass (dependent on scale normalization).
5. Δm_{21}^2 : Already measured to $< 2\%$ precision, with 1.7% deviation from our prediction—future improvements in precision can further test this.

7.2 Open Problems

- The exponent $n = 3$ is now derived (Theorem 5.5): Hecke cycling on the doublet space with multiplicative accumulation gives $M_R \propto V_k^3$.
- Second-breaking correction to θ_{23} : TBM gives 45° , experiment 47.6° .
- The 11% deviation in θ_{13} .
- A BC-dynamical derivation of the absolute mass scale.

7.3 Summary

Quantity	BC prediction	Experiment	Deviation
$R = \Delta m_{31}^2 / \Delta m_{21}^2$	33.0	33.5	1.5%
Δm_{21}^2	7.66×10^{-5}	7.53×10^{-5}	1.7%
δ_{CP}	-90°	$\sim -90^\circ$	$\sim 0\%$
$\sin^2 \theta_{12}$	1/3	0.307	8%
$\sin^2 \theta_{23}$	1/2	0.546	8%
θ_{13}	7.6°	8.54°	11%
Koide Q	2/3	0.6667	$\sim 0\%$
Tree m_ν	0	≈ 0	—
Majorana	Yes	Unconfirmed	—

Nine neutrino parameters, all with discrepancies $< 12\%$. R and Δm_{21}^2 are independent of scale normalization.

The central idea of this paper is to reduce the neutrino mass problem to character analysis over $(\mathbb{Z}/7\mathbb{Z})^*$: the pure imaginarity of the Gauss sum of χ_3 explains the vanishing of tree-level masses, while S_3 symmetry and its two successive breakings systematically produce all observable deviations. This framework will be subjected to direct experimental tests in the coming years by JUNO [18] (mass ordering), DUNE (δ_{CP}), and CMB-S4 (Σm_ν).

A Seesaw Calculation Details

Definition of M_D . The Koide circulant matrix M_D has entries

$$(M_D)_{ij} = \xi \cdot A \cdot f_{(i-j) \bmod 3},$$

where $\xi = \sqrt{7/36}$, and

$$f_k = 1 + \sqrt{2} \cos\left(\frac{2\pi k}{3} + \delta\right), \quad \delta = -13.18^\circ.$$

Numerical computation of $R = 33.0$. With $M_R = \text{diag}(6^3, 10^3, 12^3) \times M_0$, the seesaw formula $M_\nu = M_D M_R^{-1} M_D^T$ yields eigenvalues proportional to $(5.545, 1.121, 0.578) \times 10^{-3}$. The mass-squared difference ratio is then

$$R = \frac{m_3^2 - m_1^2}{m_2^2 - m_1^2} = \frac{5.545^2 - 0.578^2}{1.121^2 - 0.578^2} = 33.0.$$

Independence of A and M_0 . Since both A (the Dirac mass scale) and M_0 (the overall Majorana scale) enter as overall multiplicative factors, they cancel in the ratio $R = \Delta m_{31}^2 / \Delta m_{21}^2$. The predicted value $R = 33.0$ depends only on the Jacobi phase $\delta = -13.18^\circ$ and the doublet product ratios $6 : 10 : 12$.

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